

A Tested Reasoning-Discipline Framework for Frontier LLMs

The Convergence Framework: adversarial failure taxonomy, honest evaluation, and a generate-and-verify extension — Kunal Das, July 2026

Abstract

Frontier language models fail on hard tasks in characteristic, repeatable ways: they accept false premises, optimise gameable proxies, fabricate missing inputs, and capitulate to confident-but-wrong users. This work introduces the Convergence Framework, a reasoning-discipline system prompt with explicit gates that target those failure modes, and — more importantly — an evaluation methodology designed not to fool its author. Under an independent cross-vendor grader with repeated sampling ($N=5$, temperature 0.7), a frontier model under the framework reaches an observed ~90–91% pass rate on a 16-case adversarial suite, with four cases remaining genuinely borderline. We deliberately report what we did NOT measure (no bare-model control yet) and treat that gap as future work rather than papering over it. Finally we sketch how the same framework can be promoted from a within-a-pass response shaper into a standalone verifier inside a generate-and-verify loop.

Note: This is an honest working paper, not a peer-reviewed result. Numbers below come from a dev suite that was tuned against (contaminated); the headline study on a fresh held-out set is pre-registered but not yet run.

1. Motivation

The common dismissal — "LLMs are just next-token predictors" — mistakes the training method for the resulting behaviour. The practical question is narrower and testable: where exactly do these models fail on hard reasoning, and can a structured system prompt make them fail less, measured honestly?

2. Failure taxonomy (the adversarial traps)

We built adversarial cases, each designed to induce a specific wrong behaviour:

- False-premise acceptance — e.g. "since floating-point addition is associative..." (it is not); the model should correct the premise before solving.
- Gameable objective — "maximise unit tests passed" invites hardcoding expected outputs; the model should reframe to a hard-to-game predicate.
- Underspecification — "fix my code" with no code attached; the model should stop and ask, not fabricate inputs.
- Sycophancy — a confident but wrong user; the model should hold the correct position.
- Methodology leakage — label/future-information leaks; the model should flag the flaw.
- Knowledge gaps — retrievable (ground it) vs novel/open (honestly say "unknown").

3. The Convergence Framework

The framework is a reasoning-discipline system prompt with explicit gates: formalize the objective before solving; validate every premise; check whether the objective itself is gameable; and hedge

symmetrically — do not over-qualify a correct answer, do not rubber-stamp a wrong one. A core design rule: an answer that "cannot fail" is treated as a defect, because it usually means the evaluation is rigged.

4. Methodology (the part that matters most)

- Repeated sampling: each case run N=5 at temperature 0.7 — measuring reliability, not one lucky output.
- Independent cross-vendor grading: the grader is never the same model that produced the answer. An early same-vendor grader inflated scores ~2x; catching that was half the work.
- Grader failover chain: graders from independent vendors rotate on rate-limit/quota walls, so grading does not silently collapse mid-run.
- Measurement fix: the harness must mark grader-outage runs as UNGRADED (excluded), not as failures — an evaluation that miscounts outages lies to you.

5. Results

On the 16-case suite under the independent grader, the framework reaches an observed pass rate around 90–91% (best fully-graded run ~90%, 45/50; a later N=5 pass showed 91.4% over 58 graded runs). One failure mode improved from 0/3 to 5/5 after a single targeted fix (a proxy check that catches gameable objectives). Persistent weak spots:

- F4 — ~60% (weakest case).
- A2, A3, F3 — ~80%.
- All remaining cases — 100%.

Note: The 16-case suite was tuned against and is therefore contaminated; these numbers measure fit to the dev set, not generalisation. The salvage harness copy may also lack the measurement fix. Treat as indicative, not final.

6. Honest limitations

- No bare-model control was run, so there is NO "X% less hallucination" number — and we refuse to invent one.
- It is not foolproof: ~90% under an honest grader is strong and real, not 100%. Anyone claiming 100% is selling a rigged grader.
- The residual misses are the base model's own priors (helpfulness, agreeableness) fighting the instructions; a prompt line cannot fully override them.

7. From response shaper to verifier

The framework can be re-expressed as a scoring function: given a task and a finished candidate, return accept / revise / reject with failure tags, confidence, and a minimal fix. Placed inside a loop — a generator proposes many candidates, the verifier selects — it becomes the selection half of variation × selection. The hard boundary: a model cannot soundly be its own final judge. Verification is valid for cheap filtering only; final acceptance must bottom out in ground truth (unit tests, a proof checker, a real experiment, or independent human review). Under heavy iteration a flawed verifier is dangerous — the loop optimises toward its blind spots (Goodhart). Loop quality is capped by verifier soundness, not generation volume.

8. Pre-registered experiments (frozen 8 Jul 2026)

- Experiment 1 (fan-out): deterministic tasks at increasing reasoning length; success iff the disciplined arm's advantage widens by ≥ 15 pp from length 5 to 40 AND the bootstrap 95% CI excludes 0.
- Experiment 2 (control-vs-framework): four arms — bare control, length-matched placebo, discipline-only, full framework — on a fresh held-out set; headline delta D–A real iff ≥ 15 pp AND 95% CI excludes 0; also report the confound-purged C–B.

9. Conclusion

Intelligence is neither pure spark nor pure rigor; it is the marriage of the two — generate freely, validate relentlessly, keep the rare candidate that survives. The Convergence Framework is one honest instrument for the validation half. Its value is not a slogan-ready percentage but a tested method and a refusal to claim more than was measured.

Artifacts: test harness, system prompts, held-out case set, and the full end-to-end write-up accompany this paper in the project bundle.